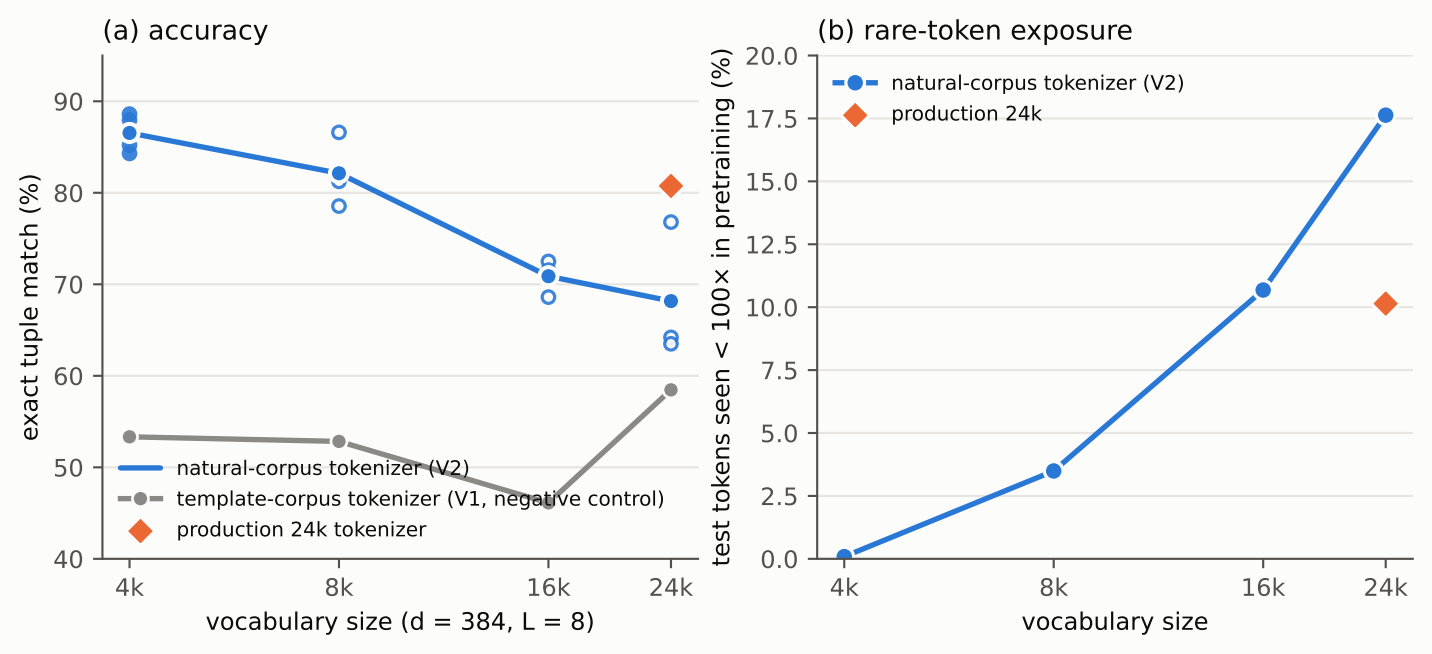
(a) accuracy
exact tuple match (%)
90
80
70
60
50
40
4k
8k
16k
24k
vocabulary size (d = 384, L = 8)
natural-corpus tokenizer (V2)
template-corpus tokenizer (V1, negative control)
production 24k tokenizer
(b) rare-token exposure
test tokens seen < 100× in pretraining (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
4k
8k
16k
24k
vocabulary size
natural-corpus tokenizer (V2)
production 24k